

MIPS SIMULATOR in Common Lisp

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Project Overview

- MIPS Simulator written in Common Lisp
- Reads MIPS assembly input file
- Parses and converts instructions to machine code
- Simulates execution of instructions
- Outputs final register and memory state



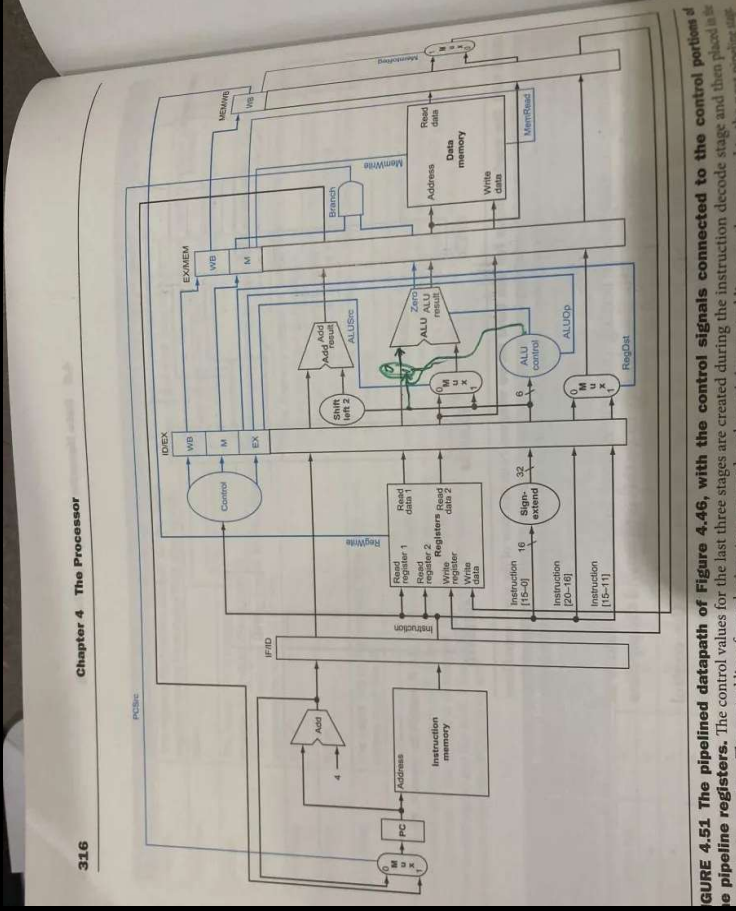
System Flow

1. Parse-assembly
2. Encode
3. Decode
4. Execution
5. Output

Key Functions

- `normalize()`:
 - cleans assembly input
- `parse-assembly()`:
 - splits instruction
- `encode()`:
 - to machine code
- `decode()`:
 - extracts instruction fields
- `register/instruction tables`
 - map registers/instructions

Design Inspiration



MIPS Pipeline Datapath Reference

Demo Overview

- Arithmetic Instructions (input-arith)

- addi, add, sub

- Logic & Shift Instructions (input-logic-shift)

- and, or, sll, srl

Focused testing of core instruction categories
(Executed using SBCL in terminal)

Testing Strategy

- Separate input files for each instruction group
 - input-arith
 - input-logic-shift
- Unit testing with test.lisp
- Verified correct instruction behavior

Challenges

- Encoding instruction to binary
- Changing from single path execution to pipeline
- Learning how to use LISP and REPL



Conclusion

- Built a MIPS simulator in Lisp
- Demonstrates instruction processing workflow
- Structured testing ensured correctness



QUESTIONS?

